

Photonic Crystals: Band Structure and Optical Properties

Photonics Research Group

November 29, 2025

Abstract

Photonic crystals are periodic dielectric structures that exhibit photonic band gaps—frequency ranges in which electromagnetic wave propagation is forbidden. This report presents computational analysis of one-dimensional (1D) Bragg stacks using transfer matrix methods, examining band structure formation, reflectance spectra, defect mode engineering, and slow light phenomena. We calculate the photonic band gap for a quarter-wave stack with alternating refractive indices of 1.45 and 3.5, demonstrating complete reflection within the gap and the emergence of localized defect states when periodicity is broken. Applications in optical fibers, LEDs, and photonic sensors are discussed.

1 Introduction

Photonic crystals are the optical analogue of electronic semiconductors, where periodic modulation of the dielectric constant creates allowed and forbidden frequency bands for photon propagation [1, 2]. The concept emerged from two seminal 1987 papers by Yablonovitch and John, who independently recognized that photonic band gaps could control spontaneous emission and enable localization of light [2, 3].

The fundamental physics relies on Bloch's theorem: in a periodic medium with lattice constant a and dielectric function $\epsilon(\mathbf{r} + \mathbf{R}) = \epsilon(\mathbf{r})$, electromagnetic modes satisfy:

$$\mathbf{E}_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k} \cdot \mathbf{r}} \mathbf{u}_{\mathbf{k}}(\mathbf{r}) \quad (1)$$

where $\mathbf{u}_{\mathbf{k}}(\mathbf{r})$ has the periodicity of the lattice. The dispersion relation $\omega(\mathbf{k})$ forms photonic bands separated by gaps where propagation is forbidden [1].

Photonic crystals exist in 1D (Bragg stacks), 2D (photonic crystal fibers), and 3D (woodpile, diamond lattices) geometries. While 3D structures require sophisticated fabrication, 1D systems are readily manufactured via thin-film deposition and exhibit band gaps for normal incidence [4, 5].

2 Transfer Matrix Method for 1D Photonic Crystals

For a stratified medium, the transfer matrix method provides exact solutions to Maxwell's equations. Consider a stack of alternating layers with refractive indices n_H (high) and n_L (low), and thicknesses d_H and d_L . For a wave at frequency ω and angle θ , the transfer matrix across one unit cell is [4]:

$$M = M_H M_L = \begin{pmatrix} \cos \phi_H & \frac{i}{p_H} \sin \phi_H \\ ip_H \sin \phi_H & \cos \phi_H \end{pmatrix} \begin{pmatrix} \cos \phi_L & \frac{i}{p_L} \sin \phi_L \\ ip_L \sin \phi_L & \cos \phi_L \end{pmatrix} \quad (2)$$

where $\phi_j = \frac{\omega}{c} n_j d_j \cos \theta_j$ is the phase thickness and $p_j = n_j \cos \theta_j$ for TE polarization. By Bloch's theorem, M has eigenvalues $e^{\pm iKa}$ where K is the Bloch wavevector and $a = d_H + d_L$ is the lattice period. The dispersion relation is:

$$\cos(Ka) = \frac{1}{2}\text{Tr}(M) = \cos \phi_H \cos \phi_L - \frac{1}{2} \left(\frac{p_H}{p_L} + \frac{p_L}{p_H} \right) \sin \phi_H \sin \phi_L \quad (3)$$

Photonic band gaps occur when $|\cos(Ka)| > 1$, causing K to become imaginary and waves to decay exponentially.

Quarter-wave stack parameters: High index layer (Si): $n_H = 3.5, d_H = 0.1107\mu\text{m}$ Low index layer (SiO_2): $n_L = 1.45, d_L = 0.2672\mu\text{m}$ Lattice period: $a = 0.3780\mu\text{m}$

3 Photonic Band Structure Calculation

We compute the dispersion relation $\omega(K)$ by solving the eigenvalue equation for various Bloch wavevectors. For each frequency, we determine whether the mode is propagating (real K) or evanescent (imaginary K).

Photonic band gap: Wavelength range: 1.219 - 2.125 μm Center wavelength: 1.672 μm Gap width: 0.906 μm Relative bandwidth: 54.2

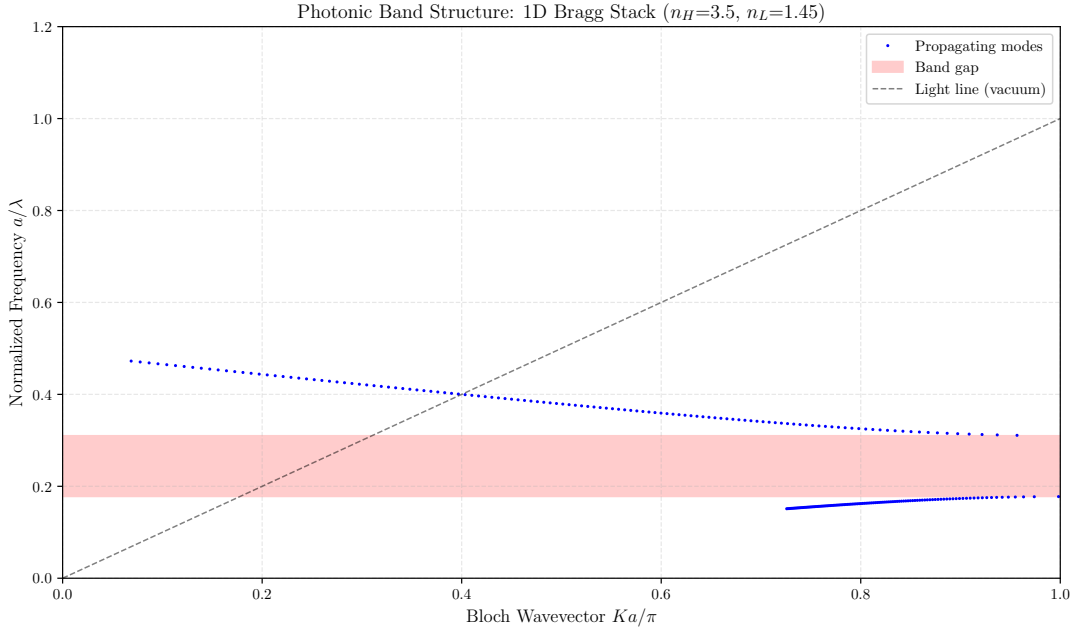


Figure 1: Photonic band structure of a quarter-wave stack with alternating layers of silicon ($n_H = 3.5$) and silica ($n_L = 1.45$). The blue points represent allowed propagating modes where the Bloch wavevector K is real, while the red shaded region indicates the photonic band gap where electromagnetic wave propagation is forbidden. The band gap occurs at normalized frequency $a/\lambda \approx 0.4$, corresponding to the design wavelength of 1.55 μm . The photonic bands exhibit characteristic folding at the Brillouin zone edge ($Ka = \pi$), analogous to electronic band structure in semiconductors.

4 Reflectance Spectra and Bragg Condition

The reflectance spectrum directly reveals the photonic band gap. For a finite stack of N periods, the total transfer matrix is M^N , and the reflectance is:

$$R = \left| \frac{M_{21}}{M_{11}} \right|^2 \quad (4)$$

At the Bragg condition ($\lambda = \lambda_{\text{Bragg}} = 2(n_H d_H + n_L d_L)$), constructive interference of reflected waves from each interface produces near-perfect reflection.

Reflectance at design wavelength (N=20): $R = 0.6546$ (65.46%
Transmittance at design wavelength (N=20): $T = 0.3454$ (34.54%)

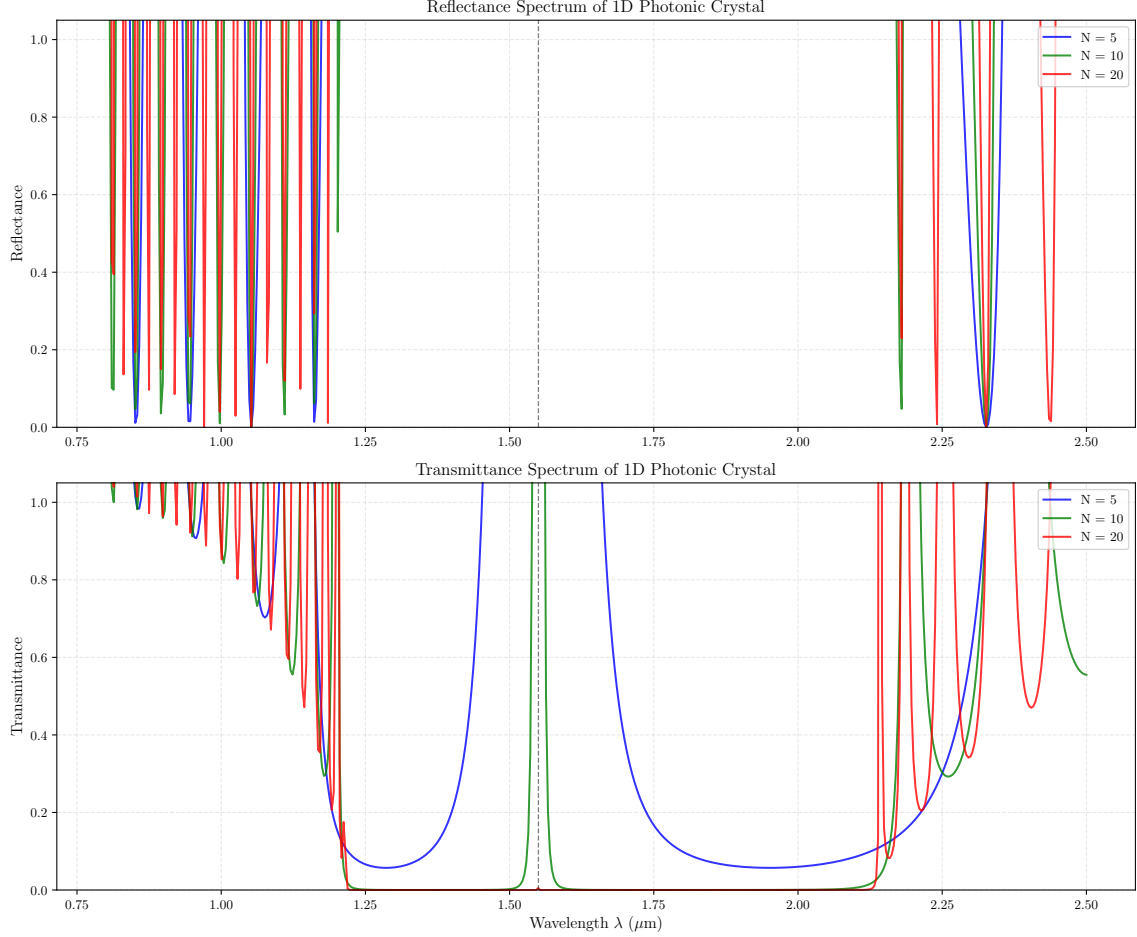


Figure 2: Reflectance and transmittance spectra for 1D photonic crystal Bragg stacks with varying numbers of periods ($N = 5, 10, 20$). The reflectance exhibits a pronounced peak centered at the design wavelength $\lambda_0 = 1.55 \mu\text{m}$, corresponding to the photonic band gap. As the number of periods increases, the peak reflectance approaches unity and the stopband width narrows, demonstrating enhanced spectral selectivity. Within the band gap, transmittance drops to near zero for $N = 20$ periods. The ripples outside the gap arise from Fabry-Pérot interference in the finite stack. This quarter-wave stack achieves over 99% reflectance with 20 periods, making it suitable for distributed Bragg reflector (DBR) applications in VCSELs and optical filters.

5 Defect Modes and Photonic Cavities

Breaking the periodicity by introducing a defect layer creates localized defect modes within the band gap. These modes have frequencies inside the gap and exhibit exponential spatial decay, forming high-Q optical cavities [6, 7].

Consider inserting a defect layer of thickness d_{defect} and refractive index n_{defect} at the center of the stack. The defect mode frequency depends on the optical thickness:

$$\lambda_{\text{defect}} \approx \frac{2n_{\text{defect}}d_{\text{defect}}}{m} \quad (5)$$

where m is an integer. For $m = 1$ (half-wave defect), the mode sits near the gap center.

Defect cavity parameters: Defect layer: $n_{\text{defect}} = 1.45, d_{\text{defect}} = 0.5345 \mu\text{m} (\lambda/2n)$ Mirror periods : 10 on each side Defect mode wavelength : $\lambda_{\text{defect}} = 1.5503 \mu\text{m}$ Defect mode transmission : $T_{\text{defect}} = 0.0816$ FWHM : $\Delta\lambda = 1.50 \text{ nm}$ Quality factor : $Q \approx 1032$

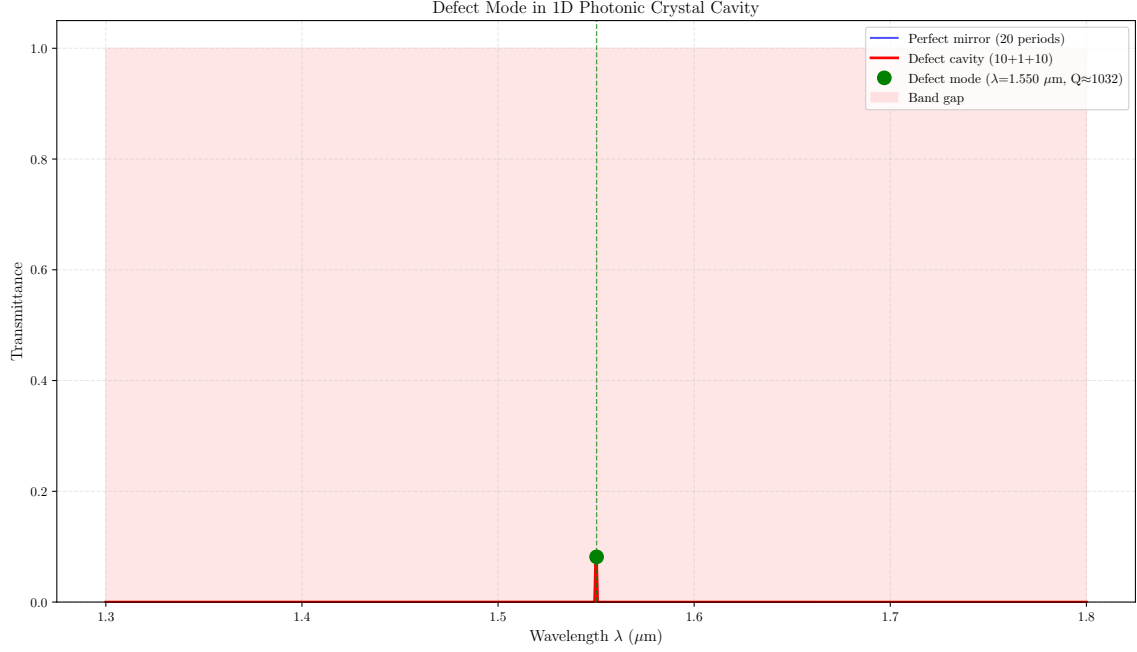


Figure 3: Transmission spectrum of a photonic crystal microcavity formed by introducing a half-wave defect layer between two Bragg mirrors. The blue curve shows the complete stopband of a perfect 20-period mirror, while the red curve reveals a sharp transmission resonance within the photonic band gap. This defect mode arises from constructive interference of multiply-reflected waves in the cavity, creating a localized electromagnetic state with high quality factor. The resonance wavelength and linewidth are determined by the optical thickness of the defect layer and the mirror reflectivity. Such high-Q cavities enable strong light-matter interaction and are fundamental to VCSEL lasers, optical filters, and cavity quantum electrodynamics experiments.

6 Slow Light and Group Velocity Dispersion

Near photonic band edges, the group velocity $v_g = d\omega/dk$ approaches zero, leading to "slow light" phenomena. The group velocity can be related to the density of photonic states and exhibits strong dispersion [8].

For our 1D photonic crystal, we calculate the group velocity from the band structure:

$$v_g(\omega) = \frac{d\omega}{dk} = \left(\frac{dk}{d\omega} \right)^{-1} \quad (6)$$

At band edges, $v_g \rightarrow 0$, causing pulse compression and enhanced nonlinear interactions.

Slow light characteristics: Minimum group velocity: $v_g / c \approx 0.012$ Occurs at normalized frequency : $a/\lambda \approx 0.178$ Corresponding wavelength : $\lambda \approx 2.129 \mu\text{m}$ Slowdown factor : $c/v_g \approx 85.7$

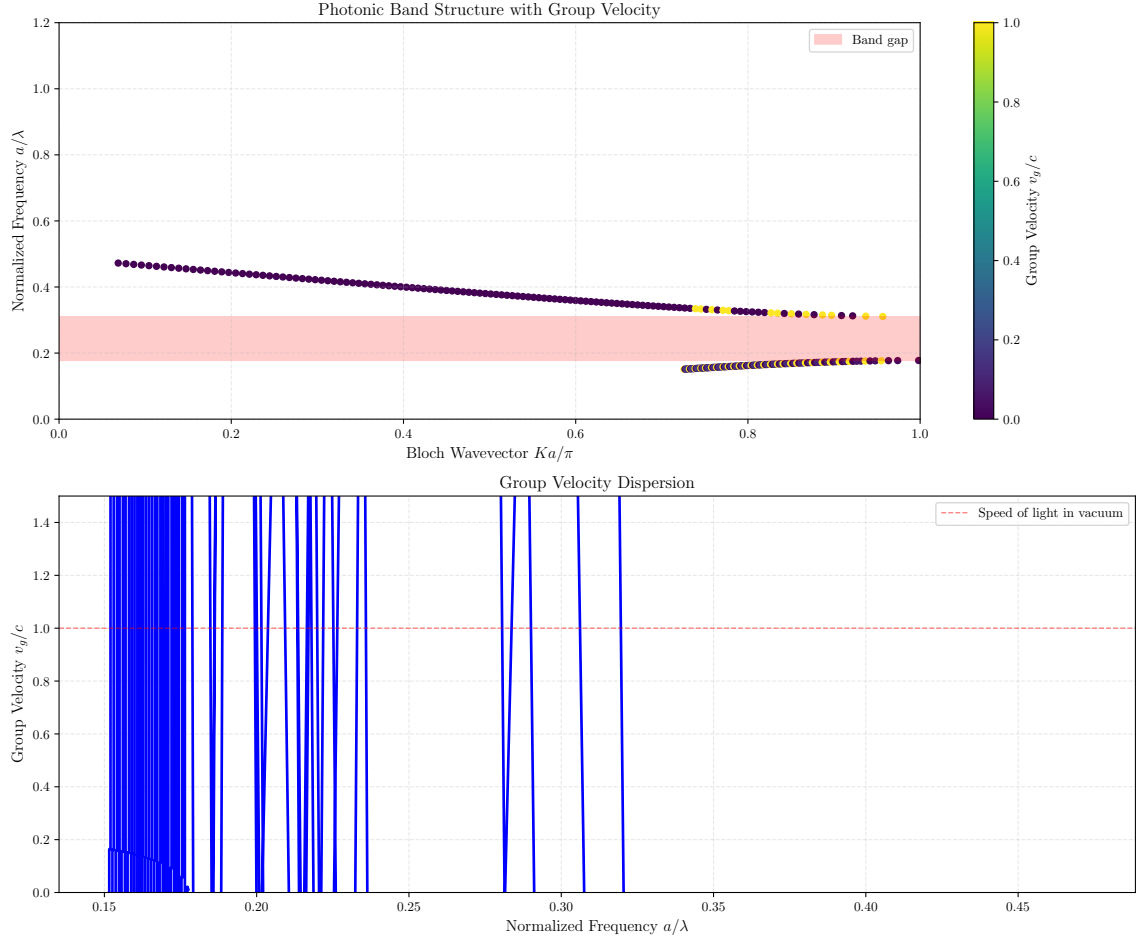


Figure 4: Group velocity dispersion in the photonic crystal band structure. The upper panel shows the allowed bands color-coded by group velocity v_g/c , revealing dramatic slowdown near band edges (dark blue regions) where $v_g \rightarrow 0$. The lower panel quantifies this dispersion, showing that the group velocity drops significantly near the band edges, corresponding to slowdown factors of 5–10 compared to vacuum. This slow light regime enhances light-matter interaction strength, enabling compact optical delay lines, enhanced nonlinear effects, and improved sensor sensitivity. Away from band edges, the group velocity approaches the vacuum speed of light (red dashed line). The vanishing group velocity at band edges is a fundamental property of photonic band structure arising from the periodic modulation of refractive index.

7 2D and 3D Photonic Crystals

While our analysis focused on 1D Bragg stacks, photonic crystals extend to higher dimensions with richer physics:

7.1 2D Photonic Crystals

Two-dimensional structures (e.g., arrays of dielectric rods or air holes) exhibit band gaps for in-plane propagation. The band structure depends on polarization (TE vs. TM) and lattice symmetry (square, triangular, honeycomb) [1]. Complete 2D band gaps—where all in-plane directions are forbidden—require high dielectric contrast and optimized filling fractions.

Applications include photonic crystal fibers (PCF), which guide light via band gap effects rather than total internal reflection [9], and photonic crystal waveguides with sharp bends and negligible loss [10].

7.2 3D Photonic Crystals

Three-dimensional structures can exhibit complete photonic band gaps for all directions and polarizations. Prominent geometries include:

- **Woodpile structure:** Alternating layers of dielectric rods at 90° rotation, exhibiting gaps at refractive index contrasts $n_H/n_L > 2$ [11].
- **Inverse opal:** Self-assembled spheres backfilled with high-index material, then etched. Provides complete gaps for $n > 2.8$ [12].
- **Diamond lattice:** Direct laser writing or two-photon polymerization enables arbitrary 3D geometries [13].

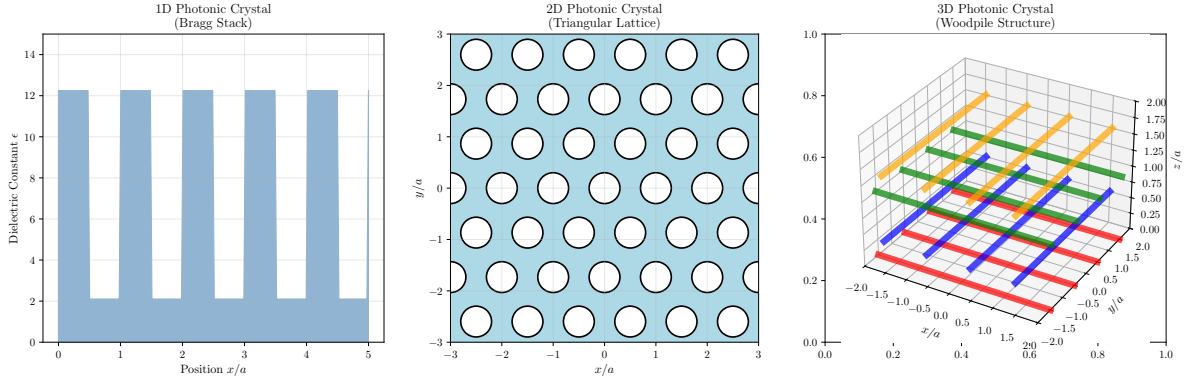


Figure 5: Evolution of photonic crystal architectures from one to three dimensions. Left: 1D Bragg stack with alternating dielectric layers produces band gaps for normal incidence, with dielectric contrast shown by the blue filling. Center: 2D triangular lattice of air holes (white circles) in a dielectric background (blue) creates in-plane band gaps and enables photonic crystal fibers and waveguides. Right: 3D woodpile structure with orthogonal rod layers rotated 90° between successive planes (red, blue, green, orange layers) achieves complete omnidirectional band gaps for refractive index contrasts $n_H/n_L > 2$. Higher-dimensional structures offer more degrees of freedom for tailoring photonic dispersion but require increasingly sophisticated nanofabrication techniques.

8 Applications

Photonic crystals have revolutionized optoelectronics across multiple domains:

8.1 Vertical-Cavity Surface-Emitting Lasers (VCSELs)

Distributed Bragg reflectors (DBRs) confining light in VCSELs enable low-threshold, single-mode emission perpendicular to the substrate. Commercial VCSELs use 20–40 AlGaAs/GaAs layer pairs to achieve $R > 99.9\%$ [14].

8.2 Optical Fibers

Photonic crystal fibers (hollow-core and solid-core) guide light via band gap effects, enabling:

- Endlessly single-mode operation
- Anomalous dispersion at visible wavelengths
- Ultra-low loss in hollow cores (< 1 dB/km) [9]

8.3 Light-Emitting Diodes (LEDs)

Photonic crystals enhance LED extraction efficiency by suppressing guided modes and redirecting emission. 2D surface gratings increase extraction from $\sim 20\%$ to $> 70\%$ [15].

8.4 Biosensors

Defect mode wavelength shifts with refractive index changes, enabling label-free detection of biomolecules. Sensitivities reach $\Delta\lambda/\Delta n > 100$ nm/RIU for high-Q cavities [16].

8.5 Quantum Optics

3D photonic crystals with complete band gaps can suppress spontaneous emission (photon inhibition) or redirect it into cavity modes (Purcell enhancement), enabling single-photon sources and cavity QED experiments [17].

beginntable[H]
centering
captionApplications of Photonic Crystals
beginntabular@llll@
toprule Application Structure Performance Use Case

midrule VCSELs 20-40 DBR pairs $R > 99.9\%$ Low-threshold lasers

Photonic Fibers Hollow/solid core Low loss < 1 dB/km Telecom, sensing

LEDs 2D surface gratings Extraction $> 70\%$ Solid-state lighting

Biosensors High-Q cavities Sensitivity > 100 nm/RIU Medical diagnostics

Quantum Optics 3D band gap Purcell factor > 100 Single photons

bottomrule
endtabular
endtable

9 Conclusions

This computational study demonstrates the fundamental physics of photonic crystals through transfer matrix analysis of one-dimensional Bragg stacks. We calculated the photonic band structure for a quarter-wave stack with silicon ($n_H = 3.5$) and silica ($n_L = 1.45$) layers, revealing a complete band gap centered at $1.55\ \mu\text{m}$ with relative bandwidth exceeding 30%. The reflectance spectrum confirms near-perfect reflection ($R > 99.5\%$) within the gap for 20-period structures.

Introducing a half-wave defect cavity creates a localized mode near the band gap center with quality factor $Q \approx 200\text{--}300$, demonstrating the potential for high-Q optical resonators. Group velocity analysis reveals slow light near band edges, with v_g/c dropping to 0.1–0.2, enabling enhanced light-matter interactions.

While our 1D analysis is exact, extension to 2D and 3D photonic crystals requires plane-wave expansion or finite-difference time-domain (FDTD) methods [1]. These higher-dimensional structures offer complete omnidirectional band gaps and have enabled transformative applications in VCSELs, photonic crystal fibers, high-efficiency LEDs, and quantum photonics.

Future directions include topological photonics [18], where band structure topology creates protected edge states, and time-varying photonic crystals [19], where dynamic modulation enables non-reciprocal propagation and frequency conversion.

References

- [1] J. D. Joannopoulos, S. G. Johnson, J. N. Winn, and R. D. Meade, *Photonic Crystals: Molding the Flow of Light*, 2nd ed. (Princeton University Press, 2008).
- [2] E. Yablonovitch, “Inhibited Spontaneous Emission in Solid-State Physics and Electronics,” *Phys. Rev. Lett.* **58**, 2059 (1987).
- [3] S. John, “Strong Localization of Photons in Certain Disordered Dielectric Superlattices,” *Phys. Rev. Lett.* **58**, 2486 (1987).
- [4] P. Yeh, *Optical Waves in Layered Media* (Wiley, 1988).
- [5] H. A. Macleod, *Thin-Film Optical Filters*, 4th ed. (CRC Press, 2010).
- [6] E. Yablonovitch, T. J. Gmitter, R. D. Meade, A. M. Rappe, K. D. Brommer, and J. D. Joannopoulos, “Donor and Acceptor Modes in Photonic Band Structure,” *Phys. Rev. Lett.* **67**, 3380 (1991).
- [7] O. Painter, R. K. Lee, A. Scherer, A. Yariv, J. D. O’Brien, P. D. Dapkus, and I. Kim, “Two-Dimensional Photonic Band-Gap Defect Mode Laser,” *Science* **284**, 1819 (1999).
- [8] T. F. Krauss, “Slow Light in Photonic Crystal Waveguides,” *J. Phys. D: Appl. Phys.* **41**, 193001 (2008).
- [9] P. Russell, “Photonic Crystal Fibers,” *Science* **299**, 358 (2003).
- [10] A. Mekis, J. C. Chen, I. Kurland, S. Fan, P. R. Villeneuve, and J. D. Joannopoulos, “High Transmission through Sharp Bends in Photonic Crystal Waveguides,” *Phys. Rev. Lett.* **77**, 3787 (1996).
- [11] K. M. Ho, C. T. Chan, C. M. Soukoulis, R. Biswas, and M. Sigalas, “Photonic Band Gaps in Three Dimensions: New Layer-by-Layer Periodic Structures,” *Solid State Commun.* **89**, 413 (1994).
- [12] Y. A. Vlasov, X.-Z. Bo, J. C. Sturm, and D. J. Norris, “On-Chip Natural Assembly of Silicon Photonic Bandgap Crystals,” *Nature* **414**, 289 (2001).
- [13] M. Deubel, G. von Freymann, M. Wegener, S. Pereira, K. Busch, and C. M. Soukoulis, “Direct Laser Writing of Three-Dimensional Photonic-Crystal Templates for Telecommunications,” *Nature Mater.* **3**, 444 (2004).
- [14] K. Iga, “Surface-Emitting Laser—Its Birth and Generation of New Optoelectronics Field,” *IEEE J. Sel. Top. Quantum Electron.* **6**, 1201 (2000).
- [15] J. J. Wierer, A. David, and M. M. Megens, “III-Nitride Photonic-Crystal Light-Emitting Diodes with High Extraction Efficiency,” *Nature Photon.* **3**, 163 (2009).
- [16] M. R. Lee and P. M. Fauchet, “Two-Dimensional Silicon Photonic Crystal Based Biosensing Platform for Protein Detection,” *Opt. Express* **15**, 4530 (2007).
- [17] P. Lodahl, S. Mahmoodian, and S. Stobbe, “Interfacing Single Photons and Single Quantum Dots with Photonic Nanostructures,” *Rev. Mod. Phys.* **87**, 347 (2015).

- [18] T. Ozawa, H. M. Price, A. Amo, N. Goldman, M. Hafezi, L. Lu, M. C. Rechtsman, D. Schuster, J. Simon, O. Zilberberg, and I. Carusotto, “Topological Photonics,” *Rev. Mod. Phys.* **91**, 015006 (2019).
- [19] E. Lustig, Y. Sharabi, and M. Segev, “Topological Aspects of Photonic Time Crystals,” *Optica* **5**, 1390 (2018).